

Lindsay S. Miles

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EDUCATION

Ph.D. in Integrative Life Sciences, Virginia Commonwealth University, Richmond, VA, USA

[2018] Dissertation Title: Patterns of Molecular Population Genetic and Phenotypic Variation Associated with Urbanization in the Western Black Widow Spider

Advisor: Dr. Brian Verrelli

Bachelor of Science in Biology, Arizona State University, Tempe, AZ, USA

[2009] GPA: 3.53 (*cum laude*)

PROFESSIONAL HISTORY

[2026 – current] Assistant Research Faculty, Center for Urban and Industrial Pest Management, Department of Entomology, Purdue University, West Lafayette, IN, USA

[2023 – 2026] Postdoctoral Researcher, Department of Entomology, Virginia Polytechnic Institute and State University, Blacksburg, VA, USA

Advisor: Dr. Warren Booth

[2021 – 2022] Adjunct Faculty, Department of Environmental Science, Portland State University, Portland, OR, USA

[2018 – 2021] Postdoctoral Fellow, Department of Biology, University of Toronto-Mississauga, Mississauga, Ontario, Canada

Advisor: Dr. Marc TJ Johnson

PROFESSIONAL CONSORTIA & AFFILIATIONS

[2023 – 2026] Postdoctoral Scholar, Fralin Life Sciences Institute at Virginia Tech, <https://fralinlifesci.vt.edu/>

[2024 – 2026] Member, Invasive Species Collaborative at Virginia Tech, <https://globalchange.vt.edu/research/working-groups/invasive-species-collaborative.html>

[2019 – Present] Member, Urban Eco Evo NET, <https://www.urbanecoevo.net/about>

[2019 – 2021] Lead Team Member, GLUE: Global Urban Evolution, <https://www.globalurbanevolution.com/>

PUBLICATIONS

[†]Co-first Authors

^{*}Undergraduate Authors

Booth W, Miles LS, Balvin O, Lewis CD, Vilimova J, Schal C, Vargo, EL. 2026. Admixture without recurrent intercontinental gene flow: contrasting patterns of genetic diversity and knockdown resistance profiles reveal independent evolution in resurgent common bed bug populations. *Journal of Pest Science* 99(121). <https://doi.org/10.1007/s10340-026-02089-z>

Carlen EJ[†], Miles LS[†], Aviles-Rodriguez K, & Booth W. 2026. A Beginner's How-To Guide to Urban Population Genetics and Genomics. *Ecology and Evolution* 16(4), e73372.

Alberti M, Grimm NB, Palkovacs EP, Urban MC, Verrelli BC, Anderson P, ... Miles LS... & Zhou, Y. 2026. Evolving nature-based solutions for urban resilience. *Science* 392(6799), 701-708. doi: 10.1126/science.aea9563

Booth W, Montgomery WI, Miles LS, Montgomery SH, Harrod C, Schunke A, Prodohl PA. 2025. Habitat mosaic limits gene flow and promotes morphological adaptation in a generalist mammal. *Ecology and Evolution* 15.12, e72588. <https://doi.org/10.1002/ece3.72588>

Miles LS, Verrelli BC, Adams R, Francioli YZ, Card DC, Balvin O, Castoe TA, Booth W. 2025. Were bed bugs the first urban pest insect? Genome-wide patterns of bed bug demography mirror global human expansion. *Biology Letters* 21, 20250061. <https://doi.org/10.1098/rsbl.2025.0061>

- Carlen, EJ... **Miles LS**, et al. 2025. Legacy effects of religion, politics, and war on urban evolutionary biology. *Nature Cities*. <https://doi.org/10.1038/s44284-025-00249-3>
- Block CJ, **Miles LS**, Lewis CD, Schal C, Vargo EL, Booth W. 2025. First evidence of the A302S Rdl insecticide resistance mutation in populations of the bed bug, *Cimex lectularius* (Hemiptera: Cimicidae) in North America. *Journal of Medical Entomology*, 1-5. doi: 10.1093/lme/tjaf033
- Miles LS**[‡], Carlen EJ[‡], Nassrullah Z[‡], Munshi-South J, Johnson MTJ. 2025. No detectable effect of urbanization on genetic drift or gene flow in specialist herbivorous insects of milkweed. *PLoS One* 20(2), e0318956. doi: [10.1371/journal.pone.0318956](https://doi.org/10.1371/journal.pone.0318956)
- Miles LS**, Adams R, Franciolo YZ, Card DC, Castoe TA, Booth W. 2024. A chromosome-level reference genome for the common bed bug, *Cimex lectularius*, with identification of sex chromosomes. *Journal of Heredity*, esae071. doi: 10.1093/jhered/esae071
- AM Savage, MJ Willmott, PM Garcia, Z Jagiello, D Li, A Malesis, **LS Miles**, ..., KM Gotanda. 2024. Online toolkits for collaborative and inclusive research in urban evolutionary *Ecology and Evolution*, 14(6), 1-18. doi: [10.1002/ece3.11633](https://doi.org/10.1002/ece3.11633)
- Miles LS**[‡], Waterman H[‡], Ayoub NA, Garb JE, Haney RA, Rosenberg MS, Krabbenhoft T, Verrelli BC. 2024. Insight into the adaptive role of arachnid genome-wide duplication through chromosome-level genome assembly of the Western black widow spider. *Journal of Heredity*, 115(3), 241-252. doi: [10.1093/jhered/esae018](https://doi.org/10.1093/jhered/esae018)
- Winchell KM, Aviles-Rodriguez K, Carlen EJ, **Miles LS**, Charmantier A, De León LF, Gotanda K, Rivkin LR, Szulkin M, Verrelli BC. 2022. Moving past the challenges and misconceptions in urban adaptation research. *Ecology and Evolution*, 12(11), e9552. <http://doi.org/10.1002/ece3.9552>
- Miles LS**[‡], Murray-Stoker D[‡], Nahn V[‡], Johnson MTJ. 2022. Effects of urbanization on specialist insect communities of milkweed are mediated by spatial and temporal variation. *Ecosphere*, 19(9), e4222. <https://doi.org/10.1002/ecs2.4222>
- Verrelli BC, Alberti M, Des Roches S, Harris NC, Hendry AP, Johnson MTJ, Savage AM, Charmantier A, Gotanda KM, Govaert L, **Miles LS**, ..., Ziter C. 2022. A global horizon scan for urban evolutionary ecology. *Trends in Ecology and Evolution*. <https://doi.org/10.1016/j.tree.2022.07.012>
- Santangelo JS, Ness RW, Cohan, Fitzpatrick CR, Innes SJ, Koch S, **Miles LS**, Munim S, PeresNeto PR, Prashad C, Tong AT,... Johnson MTJ. 2022. Global urban environmental change drives adaptation in white clover. *Science*, 375 (6586), 1275-1281. <https://doi.org/10.1126/science.abk0989>
- Tassone E[‡], **Miles LS**[‡], Rosenberg MS, Dyer RJ, Verrelli BC. 2021. Evolutionary stability, landscape heterogeneity, and human land-use shape population genetic connectivity in the Cape Floristic Region biodiversity hotspot. *Evolutionary Applications*, 14(4), 1109-1123. <https://doi.org/10.1111/eva.13185>
- Miles LS**, Carlen EJ, Winchell KM, Johnson MTJ. 2021. Urban evolution comes into its own: Emerging themes and future directions of a burgeoning field. *Evolutionary Applications*, 14, 3-11. <https://doi.org/10.1111/eva.13165>
- Miles LS**, Ayoub NA, Garb JE, Haney RA, & Verrelli BC. 2020. Ovarian Transcriptomic Analyses in the Urban Human Health Pest, the Western Black Widow Spider. *Genes*, 11(1), 87. <https://doi.org/10.3390/genes11010087>
- Miles LS**[‡], Rivkin LR[‡], Munshi-South J, Johnson MTJ, & Verrelli BC. 2019. Gene flow and genetic drift in urban environments. *Molecular Ecology*, 28, 4138-4151. <https://doi.org/10.1111/mec.15221>
- Miles LS**[‡], Breitbart ST[‡], Wagner HH, & Johnson MTJ. 2019. Urbanization shapes the ecology and evolution of plant-arthropod herbivore interactions. *Frontiers in Ecology and Evolution*, 7, 310. <https://doi.org/10.3389/fevo.2019.00310>
- Rivkin RL, Santangelo JS, Alberti M, Aronson MFJ, de Keyser CW, Diamond SE, Fortin MJ, Frazee LJ, Gorton AJ, Hendry AP, Liu Y, Losos JB, MacIvor JS, Martin RA, McDonnell M, **Miles LS**, ... Johnson MTJ. 2018. A roadmap for urban evolutionary change. *Evolutionary Applications*, 1-15. <https://doi.org/10.1111/eva.12734>
- Miles LS**, Dyer RJ, & Verrelli BC. 2018. Urban hubs of connectivity: Contrasting patterns of gene flow within and among cities in the Western black widow spider. *Proceedings of the Royal Society B*, 285, 20181224. <https://doi.org/10.1098/rspb.2018.1224>
- Miles LS**, Johnson JC, Dyer RJ, & Verrelli BC. 2018. Urbanization as a facilitator of gene flow in a human health pest. *Molecular Ecology*, 27(16), 3219-3230. <https://doi.org/10.1111/mec.14783>
- Johnson JC, **Miles LS**, Trubl PJ, & Hagenmaier A^{*}. 2014. Maternal effects and egg provisioning affect early development speed and clutch sex ratio, but have no effect on sibling cannibalism in the black widow spider (*Latrodectus hesperus*). *Animal Behaviour*, 91, 67-73. <https://doi.org/10.1016/j.anbehav.2014.02.031>
- Still MB^{*}, **Miles LS**, Gburek TM^{*}, & Johnson JC. 2013. Adverse effects of fluorescent dust marking on the behavior of Western black widow spiderlings. *Entomologia Experimentalis et Applicata*, 150(1), 28-31. <https://doi.org/10.1111/eea.12140>
- Johnson JC, Trubl P, & **Miles L**. 2012. Black widows in an urban desert: city-living compromises spider fecundity and egg investment despite urban prey abundance. *American Midland Naturalist*, 168(2), 333-340. <https://doi.org/10.1674/0003-0031-168.2.333>
- Trubl P, Gburek T^{*}, **Miles L** & Johnson JC. 2012. Black widows in an urban desert: population variation in an urban pest across metropolitan Phoenix. *Urban Ecosystems*, 15(3), 599-609. <https://doi.org/10.1007/s11252-011-0223-2>
- Johnson JC, Blackmore V^{*}, Trubl P, & **Miles L**. 2011. Male black widows court well-fed females more than starved females: silken cues indicate sexual cannibalism risk. *Animal Behaviour*, 82, 383-390. <https://doi.org/10.1016/j.anbehav.2011.05.018>

CONFERENCE PRESENTATIONS

- Miles LS**, Block B, Gustafuson N, DeWitt C, Jones A, Wilson M, Miller D, Booth W. *Spatio-Temporal Population Genomics and Target-Site Associated Insecticide Resistance Profiles of the German Cockroach, *Blattella germanica*, from Low Income Housing in Virginia*. National Conference on Urban Entomology, May 2026, Atlanta, GA, USA
- Miles LS**. *Bed bug genome-wide patterns of demographic history mirror human global expansion*. PopGroup 58, January 2025, Sheffield, UK.
- Miles LS**. *Bed bug genome-wide patterns of demographic history mirror human global expansion*. Evolution Conference, July 2024, Montreal, QC, Canada.
- Miles LS**. *Landscape Genetics as a Tool for Managing Human-impacted Landscapes: Closing remarks*. Symposium Co-Chair for the International Association for Landscape Ecology- North America Special Symposium: Landscape Genetics as a Tool for Managing Human-impacted Landscapes. IALE-NA Conference, May 2020, online.
- Miles LS**. *Non-adaptive urban evolution in arthropod populations*. Invited talk at the Entomology Society of America Conference in SysEB Section Symposium: What Everyone Ought to Know about Insect Biodiversity in the Urban Environment. Entomology Conference, November 2019, St. Louis, Missouri, USA.
- Miles LS**, Rivkin LR, Munshi-South J, Johnson MTJ, Verrelli BC. *Genetic drift and gene flow in urban environments*. Evolution Conference, June 2019, Providence, RI, USA.
- Miles LS**, Haney RA, Garb JE, Ayoub NA, Verrelli BC. *Urbanization shapes differential transcriptomic patterns across populations of a human health pest, the Western black widow spider*. Evolution Conference, June 2019, Providence, RI, USA.
- Miles LS**, Verrelli BC. *Urbanization facilitates gene flow among natural habitats of the Western black widow spider*. Invited talk at Evolution in Urban Ecosystem Symposium. Evolution Conference, June 2017, Portland, Oregon, USA.
- Miles LS**, Verrelli BC. *Sex in the City: Urbanization facilitates gene flow among natural habitats of the Western black widow spider*. Evolution Conference, June 2016, Austin, Texas, USA.
- Miles LS**, Verrelli BC. *A population genetic approach to investigate effects of urbanization and habitat fragmentation on the Western black widow spider, *Latrodectus hesperus**. 4th Annual Integrative Life Sciences Student Research Showcase, November 2013, Richmond, Virginia, USA.
- Miles LS**, Johnson JC, Ziemba R, Verrelli BC. *A population genetic approach to investigate effects of urbanization and habitat fragmentation on the Western black widow spider, *Latrodectus hesperus**. Animal Behavior Society Conference, July 2011, University of Indiana, Bloomington, Indiana, USA.
- Miles LS**, Johnson JC, Ziemba R, Verrelli BC. *A population genetic approach to investigate effects of urbanization and habitat fragmentation on the Western black widow spider, *Latrodectus hesperus**. Evolution Conference, June 2011, Norman, Oklahoma, USA.

INVITED SYMPOSIA & SEMINARS

- Miles LS**. *Genetic drift and gene flow in urban environments*. Ecology, Evolution, and Behaviour Seminar Series at Queens University, February 2021, Kingston, Ontario, Canada.
- Miles LS**. *Genetic drift and gene flow in urban environments*. Biology Seminar Series at the University of Guelph, November 2019, Guelph, Ontario, Canada.
- Miles LS**. *Urbanization shapes patterns of molecular evolution in the Western black widow spider*. 14th Annual Atwood Colloquium, April 2019, Toronto, ON, Canada.
- Miles LS**. *Urbanization shapes patterns of molecular evolution in the Western black widow spider*. Biology Seminar Series at the University of Toronto Scarborough, February 2019, Scarborough, ON, Canada.
- Miles LS**, Haney RA, Ayoub NA, Verrelli BC. *Differential gene expression associated with urbanization in Western black widow spiders*. NCGAS de novo Transcriptome Assembly Workshop, October 2018, Bloomington, Indiana, USA.
- Miles LS**, Verrelli BC. *Comparing urban gene flow between Southwestern cities in the Western black widow spider, *Latrodectus hesperus**. Synthesis in the City: Urban Evolutionary Ecology Workshop, July 2017, Toronto, Ontario, Canada.

WORKSHOPS

- [NSF-Funded Mountain Lake Biological Station Grant Writing Workshop](#). July 2023, Pembroke, VA, USA.
- [NSF-Funded Urban Eco-Evo Research Collaborative Network Workshop](#). February 2023, Seattle, WA, USA.
- [Society for Molecular Biology and Evolution \(SMBE\)- Funded Satellite Workshop on Urban Evolutionary and Ecological 'Omics](#). November 2022. Washington, D.C.

NSF-Funded Evo-LTER conference: [Building collaborations to investigate the effects of urbanization on eco-evolutionary dynamics across the LTER network](#). August 2022, Albuquerque, NM, USA

SEEP: [Integrating society, ecology, evolution, and plasticity to advance urban evolutionary ecology](#). May 2022, Tempe, AZ, USA.

[Synthesis in the City: Urban Evolutionary Ecology Workshop](#). July 2017, Toronto, Ontario, Canada.

FELLOWSHIPS, AWARDS, & GRANTS

[2024] Travel Award (\$600): Virginia Polytechnic and State University Postdoctoral Associates Travel Award

[2020] Symposia Funding (\$2500): University of Toronto Mississauga Centre for Urban Environments Symposia/Workshop Coordinator Grant

[2019] Research Grant (\$2000): University of Toronto Mississauga Postdoctoral Fellow Research Grant

[2018] Writing Award (\$500): VCU Graduate School Distinguished Dissertation Award

[2013] Travel Award (\$200): Integrative Life Sciences Showcase Presentation Competition

[2010] Research Award (\$4000): Arizona State University/NSF Central Arizona Phoenix Long Term Ecological Research Project

TEACHING EXPERIENCE

[Instructor](#)

[2021 – 2022] Biological Concepts and Applications; Department of Environmental Science, Portland State University

[Graduate Teaching Assistant](#)

[2015 – 2016] Capstone Molecular Biology Techniques; Department of Life Sciences, Virginia Commonwealth University

[2011] School of Life Sciences Undergraduate Research (SOLUR) Assistant, School of Life Sciences, Arizona State University

[2008 – 2013] General Biology for non-majors, General Biology I Lecture, General Biology II Laboratory, General Genetics Recitation, Organic Evolution Lecture, Organic Evolution Recitation, School of Life Science, Arizona State University

[Guest Lecturer](#)

[2024] Ecology; Department of Biology, Virginia Polytechnic and State University

[2019 – 2020] Evolutionary Biology; Department of Biology, University of Toronto Mississauga

[2017 – 2018] Evolutionary Biology; Department of Biology, Virginia Commonwealth University

[Student mentorship](#)

[2019] Research Opportunity Project; Department of Biology, University of Toronto Mississauga

GUEST EDITOR

[2020] Evolutionary Applications

MANUSCRIPT REVIEWER

Biological Journal of the Linnean Society, BMC Genetics, Bulletin of Entomological Research, Conservation Science and Practice, Discover Agriculture, Ecology, Ecology & Evolution, Evolutionary Applications, Genome Biology and Evolution, Genes, Global Change Biology, Integrative and Comparative Biology, Journal for Urban Ecology, Journal of Heredity, Molecular Ecology, Perspectives in Plant Ecology, Evolution, and Systematics, PLOS One, Proceedings of the Royal Society B, Scientific Reports, Trends in Ecology and Evolution, Urban Ecosystems

PROFESSIONAL MEMBERSHIPS

[2011 – Present] Society for the Study of Evolution

[2026 – Present] Entomological Society of America

[2011 – Present] Association for Women in Science

[2018 – 2021] Canadian Society for Ecology and Evolution

[2009 – 2012] Animal Behavior Society

ACADEMIC SERVICE

- [2023 – 2025] Virginia Tech Postdoc Association (VTPA), Virginia Polytechnic Institute and State University
Serving as President (2024-2025)
Served as Social Chair of organization (2023 – 2024)
- [2018 – 2021] University of Toronto Mississauga Postdoctoral Association, University of Toronto Mississauga
Served as Vice President of organization 2018-2019
Served as President of organization 2019-2021
- [2015 – 2017] Integrative Life Science Student Organization Board Member, Virginia Commonwealth University
Served as Vice President of student organization 2016-2017
Served as Secretary of student organization 2015-2016
- [2013 – 2015] Graduate Organization of Biology Students Member at Large, Virginia Commonwealth University

SCIENTIFIC OUTREACH

- [2024] Hokie Bug Fest at Virginia Tech
- [2019] Discovery@Riverwood, Community Lecture Series at the Riverwood Conservancy.
- [2018 – Present] Editor for Life in the City: urban evolution blog.
- [2010 – Present] Ask-A-Biologist, online.
- [2017] Commonwealth Graduate Education Day, Washington, D.C.
- [2011] The Wonderlab Museum, Bloomington, IN.
- [2010 – 2013] Arizona Science Center, Phoenix, AZ. Goosebumps: The Science of Fear.

POPULAR PRESS

- [2025] [Early Humans Settled in Cities. Bedbugs Followed Them.](#) in New York Times.
- [2025] [Bedbugs have been bugging us since before beds](#) in Science Friday.
- [2025] [Researchers find first evidence of potential bed bug insecticide resistance in gene mutation](#) in VT news.
- [2020] Darwin in Times Square, the science of urban evolution, documentary film by Sagamedia.
- [2019] [Urbanization facilitates migration of the human health pest, Western black widow spider](#) in Atlas of Science.
- [2019] [How Cities Reshape the Evolutionary Path of Urban Wildlife](#) in Wired.
- [2019] [Cities Can Serve as Cauldrons of Evolution](#) in The Scientist.
- [2018] [City Spider: How Black Widow Spiders Thrive in a Warmer, Urban Environment](#) in Phoenix New Times
- [2009] [Spider lab at ASU's West campus frightens, then educates](#) in ASU Now